

EXPERIMENT 18: STUDENT–TEACHER–SUBJECT CODE DATABASE

AIM

To write a Prolog program for a student–Teacher–Subject Code database and retrieve the relationship between students, teachers and subjects.

OBJECTIVE

- To create a relational database using Prolog facts.
- To represent student, teacher and subject information.
- To retrieve information using logical queries.
- To understand relationships between different entities.

ALGORITHM

1. Start.
2. Define student facts.
3. Define teacher facts.
4. Define subject-code facts.
5. Create a relationship between student, teacher and subject.
6. Query the database.
7. Display the matching information.
8. Stop.

SOURCE CODE

% Student - Teacher - Subject Code Database

student(ravi).

student(priya).

student(kumar).

student(anitha).

teacher(suresh).

teacher(meena).

teacher(rajesh).

subject_code(ai, 'CSA01').

subject_code(cloud, 'CSA02').

subject_code(compiler, 'CSA03').

teaches(suresh, ai).

teaches(meena, cloud).

teaches(rajesh, compiler).

studies(ravi, ai).

studies(priya, cloud).

studies(kumar, compiler).

studies(anitha, ai).

student_teacher_subject(Student, Teacher, Subject, Code) :-

studies(Student, Subject),

teaches(Teacher, Subject),

subject_code(Subject, Code).

Query

?- student_teacher_subject(ravi, Teacher, Subject, Code).

OUTPUT

```
% c:/Users/dloge/Desktop/prl/3.pl compiled 0.00 sec, 0 clauses
?- teaching_subjects(john, Subject).
Subject = math.

?- taking_students(science, Student).
Student = alice.

?- taking_students(science, Student).
Student = alice.

?-
% c:/Users/dloge/Desktop/prl/3.pl compiled 0.00 sec, 0 clauses
?- teaching_subjects(john, Subject).
Subject = math.

?- taking_students(science, Student).
Student = alice.

?- taking_students(math, Student), taking_students(art, Student).
Student = dave.
```

RESULT

Teacher = suresh,

Subject = ai,

Code = 'CSA01'.

Another query:

?- student_teacher_subject(priya, Teacher, Subject, Code).

Output:

Teacher = meena,

Subject = cloud,

Code = 'CSA02'.

The Prolog program successfully represents and retrieves Student–Teacher–Subject Code relationships.